



Pradnya Niketan Education Society, Pune

NAGESH KARAJAGI *ORCHID* COLLEGE OF ENGINEERING & TECHNOLOGY, SOLAPUR

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Final year Project-II Work Diary

Academic Year: 2025-2026

Project Title: Smart Traffic Density & Signal Optimization System

Project Group No: 07

Project Group Members: 04

Sr No.	Name	Roll No.	Exam Seat No.
1	Zahid Khan	43	2262231995045
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Project Guide: Prof. N. B. Aherwadi

Project Schedule

Month	Week	Project Activity
January	Week 4	Formation of project group
February	Week 1	Finalization of project idea
	Week 2	Finalization of project title
	Week 3	Study of existing systems
	Week 4	Literature survey
March	Week 1	Literature Review
	Week 2	Project Abstract
	Week 3	System Architecture
	Week 4	Algorithm
April	Week 1	Project Implementation
	Week 2	Class Case, Use Case, Activity, Sequence, DFD, UML Diagrams
	Week 3	Report writing, Project Work Diary
	Week 4	Report Submission

Project Progress Sheet

Semester – II

Activity week: January Week 4 From 25/01/2026 to 30/01/2026

Activity according to schedule:

1. Formation of project group

Group is formed with four members in a team.

Date: 30/01/2026

Guide: Prof. N. B. Aherwadi

Activity week: February Week 1 From 01/02/2026 to 06/02/2026

Activity according to schedule:

1. Finalization of project idea

Work done:

The selected idea was an AI-Based Smart Traffic Management System using YOLOv8, FastAPI, OpenCV, and React for real-time vehicle detection and dynamic traffic signal management.

Date: 06/02/2026

Guide: Prof. N. B. Aherwadi

Activity week: February Week 2 From 08/02/2026 to 13/02/2026

Activity according to schedule:

1. Finalization of project title

Based on the selected idea and research, the final project title was decided as “Smart Traffic Density & Signal Optimization System.”

Date: 13/02/2026

Guide: Prof. N. B. Aherwadi

Activity week: February Week 3 From 15/02/2026 to 20/02/2026

Activity according to schedule:

1. Study of existing systems

Work done:

A study of existing traffic management and monitoring systems was conducted, including traditional fixed-timer traffic signals and AI-based traffic analysis systems. Their limitations such as inefficient signal timing, traffic congestion, lack of real-time optimization, and limited scalability were identified.

Date: 20/02/2026

Guide: Prof. N. B. Aherwadi

Activity week: February Week 4 From 22/02/2026 to 27/02/2026

Activity according to schedule:

1. Literature survey

Work done:

A detailed literature survey was carried out on AI-based traffic management systems, YOLOv8 object detection, OpenCV video processing, FastAPI-based REST APIs, real-time traffic monitoring, and dynamic traffic signal optimization techniques used in smart transportation systems.

Date: 27/02/2026

Guide: Prof. N. B. Aherwadi

Activity week: March Week 1 From 01/03/2026 to 06/03/2026

Activity according to schedule:

1. Literature Review

Work done:

Comprehensive literature review writing began, covering AI-based traffic monitoring systems, YOLOv8 vehicle detection models, OpenCV video processing, real-time traffic analysis, and dynamic signal optimization techniques.

Date: 06/03/2026

Guide: Prof. N. B. Aherwadi

Activity week: March Week 2 From 08/03/2026 to 13/03/2026

Activity according to schedule:

1. Project Abstract

Work done:

It included the project objective, methodology, architecture, and expected outcome. The abstract described how the system would detect and count vehicles using YOLOv8, process traffic videos through a FastAPI backend, and dynamically optimize traffic signals using a real-time monitoring dashboard.

Date: 13/03/2026

Guide: Prof. N. B. Aherwadi

Activity week: March Week 3 From 15/03/2026 to 20/03/2026

Activity according to schedule:

1. Project Analysis

Work done:

The team designed the workflow: Traffic Video Input → YOLOv8 Vehicle Detection → FastAPI Backend → OpenCV Frame Processing → Real-Time Statistics → React Dashboard → Dynamic Signal Optimization. A high-level architecture diagram and system workflow were prepared.

Date: 20/03/2026

Guide: Prof. N. B. Aherwadi

Activity week: March Week 4 From 22/03/2026 to 27/03/2026

Activity according to schedule:

1. System Architecture

Work done:

Algorithm design for the project included vehicle detection using YOLOv8, video frame processing with OpenCV, real-time traffic density calculation, dynamic signal timing logic, API request handling, and live dashboard update mechanisms.

Date: 27/03/2026

Guide: Prof. N. B. Aherwadi

Activity week: April Week 1 From 01/04/2026 to 04/04/2026

Activity according to schedule:

1. Algorithm

Work done:

The FastAPI backend was developed using APIs for real-time traffic monitoring and dashboard communication. YOLOv8 vehicle detection, OpenCV video processing, Reinforcement Learning-based signal optimization, and JSON-based real-time analytics were successfully implemented and tested.

Date: 04/04/2026

Guide: Prof. N. B. Aherwadi

Activity week: April Week 2 From 05/04/2026 to 10/04/2026

Activity according to schedule:

1. Class Case, Use Case, Activity, Sequence, DFD, UML Diagrams

Work done

Prepared Class Diagram, Use Case Diagram, Activity Diagram, Sequence Diagram, DFD, and UML Diagrams based on the Smart Traffic Density & Signal Optimization System architecture, workflow, vehicle detection process, and dynamic signal management functionalities.

Date: 10/04/2026

Guide: Prof. N. B. Aherwadi

Activity week: April Week 3 From 12/04/2026 to 17/04/2026

Activity according to schedule:

1. Report writing

2. Work Diary

Work done:

Prepared the project report and updated the work diary based on all completed tasks including literature review, system architecture, YOLOv8 vehicle detection, FastAPI backend development, React dashboard implementation, and dynamic traffic signal optimization.

Date: 17/04/2026

Guide: Prof. N. B. Aherwadi

Activity week: April Week 4 From 20/04/2026 to 29/04/2026

Activity according to schedule:

Report Submission

Work done:

Finalized and submitted the project report after completing all required documentation and reviews.

Date: 29/04/2026

Guide: Prof. N. B. Aherwadi

(Prof. G. N. Chanderki)

(Prof. M. S. Otari)

Project Coordinator

HOD